

Conn_02x16_Odd_Even

Pin	Signal
PE2	SENSE_RESV2
2	SENSE_RESV3
+3V3	
4	
+5V	GND
6	A8
PE3	A9
8	
PE15	
10	A11
T1C3	
12	A10
T1C4	
14	A11
T1C2	
16	A12
T1C3N	
18	A12
T1C1	A13
20	
T1C2N	A14
22	
	A15
24	
GND	A14
T1C1N	
26	A15
+5V	
28	
+3V3	GND
30	
I2C2_SDA	I2C2_SCL
32	

Front Aux

Pin	Signal	Notes
1	I2C2_SCL	
2	GND	
3	I2C2_SDA	
4	+3V3	
5	A10	
6	+5V	
7	A11	
8	A12	
9	GND	
10	LPTIM2/A1	
11	LPTIM2/A2	
12	LPTIM2/A4	
13	A5	
14	PA1	Potential USART4
15		
16		

The image shows two pin connection diagrams. The left diagram, labeled 'Conn_01x04_Pin', shows a 4-pin connector with the following connections: Pin 1 is +3V3, Pin 2 is I2C2_SDA, Pin 3 is I2C2_SCL, and Pin 4 is GND. The right diagram, labeled 'Conn_01x06_Pin', shows a 6-pin connector with the following connections: Pin 1 is +3V3, Pin 2 is SPI2_NSS/PB12, Pin 3 is SPI2_CLK/PB13, Pin 4 is SPI2_MISO/PB14, Pin 5 is SPI2_MOSI/PB15, and Pin 6 is GND. Both diagrams include a 'J6' label next to the connector pins.

EN_3	5	6	MOTOR4_B
FWD_4	7	8	CTRL_EXT_LOAD2
EN_4	9	10	USART2_RX/PD6
USART2_TX/PD5	11	12	EXT_LOAD2
BATT	13	14	PB4 Charge Current
I2C1_SCL/PB8	15	16	PD7 RESV
+3V3	17	18	I2C1_SDA/PB9
+5V	19	20	GND
HV	21	22	EXT_LOAD1
FWD_2	23	24	GND
EN_2	25	26	CTRL_EXT_LOAD1
EN_1	27	28	MOTOR3_A
FWD_1	29	30	MOTOR3_B
MOTOR1_A	31	32	MOTOR1_B

Pin-to-pin mapping for the STM32F405VGT6 to the U1 mm_mcu_v0 module. The diagram shows two columns of pins with their corresponding functions and pin numbers. A central block represents the module with pins 1 through 75. The left column lists pins 1 through 24, and the right column lists pins 25 through 75. The top of the diagram shows power and ground connections: HV to pin 10 (GND), +5V to pin 2 (GND), and 3V3 to pins 31, 33, 48, and 50 (GND).

Pin	Function	Pin	Function
1	XSHUT1	25	PE1
2		26	TSC_G7_I03/PE4
3	FWD_1	27	TSC_G7_I04/PE5
4	EN_1	28	TSC_G7_I01/PE2
5	EN_2	29	TSC_G7_I02/PE3
6	FWD_2	30	NRST
7	T3C1/PC6	31	PE0
8	T3C4/PC9	32	I2C1_SCL/PB8
9	T4C3/PD14	33	I2C1_SDA/PB9
10	GND	34	USART1_RX/PB7
11	GND	35	USART1_TX/PB6
12	T4C1/PD12	36	PD7
13	T4C2/PD13	37	PB4
14	T4C3/PD14	38	USART2_TX/PD5
15	T4C4/PD15	39	USART2_RX/PD6
16	USART3_TX/PD8	40	PD3
17	USART3_RX/PD9	41	PD4
18		42	PC11
19		43	PD2
20	PD10	44	XSHUT4 PC1
21	PD11	45	PC10
22	PD1	46	CAN1_TX/PD1
23	T3C1/PC6	47	PA15
24	T3C2/PC7	48	CAN1_RX/PD0
25	T3C3/PC8	49	
26	T3C4/PC9	50	
27		51	
28		52	
29		53	
30		54	
31		55	
32		56	
33		57	
34		58	
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37		61	
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39		63	
40		64	
41		65	
42		66	
43		67	
44		68	
45		69	
46		70	
47		71	
48		72	
49		73	
50		74	
51		75	

Pin connection diagram for J12 connector:

Pin	Signal
1	J5_Conn01x02_Pin (HV)
2	GND
3	GND
4	PD10
5	HV
6	PD10
7	+5V
8	+5V
9	USART3_TX/PD8
10	USART3_RX/PD9
11	BATT
12	BATT
13	EXT_LOAD2
14	GND
15	MOTOR4_B
16	MOTOR4_A

J12

Conn_02x08_Odd_Even		
3_A	1	2 MOTOR3_B
AD1	3	4 GND
V	5	6 PD10
V	7	8 +5V
PD8	9	10 USART3_RX/PD9
BATT	11	12 BATT
AD2	13	14 GND
4_B	15	16 MOTOR4_A

Diagram showing the pin connections for headers J8, J9, and J10. Each header has 6 pins. The connections are as follows:

- J8 Conn_01x06_Pin:**
 - Pin 1: XSHUT1
 - Pin 2: I2C2_SDA
 - Pin 3: I2C2_SCL
 - Pin 4: GND
 - Pin 5: +3V3
- J9 Conn_01x06_Pin:**
 - Pin 1: XSHUT1
 - Pin 2: I2C2_SDA
 - Pin 3: I2C2_SCL
 - Pin 4: GND
 - Pin 5: +3V3
- J10 Conn_01x06_Pin:**
 - Pin 1: XSHUT1
 - Pin 2: I2C2_SDA
 - Pin 3: I2C2_SCL
 - Pin 4: GND
 - Pin 5: +3V3

J17
Conn_01x06_Pin

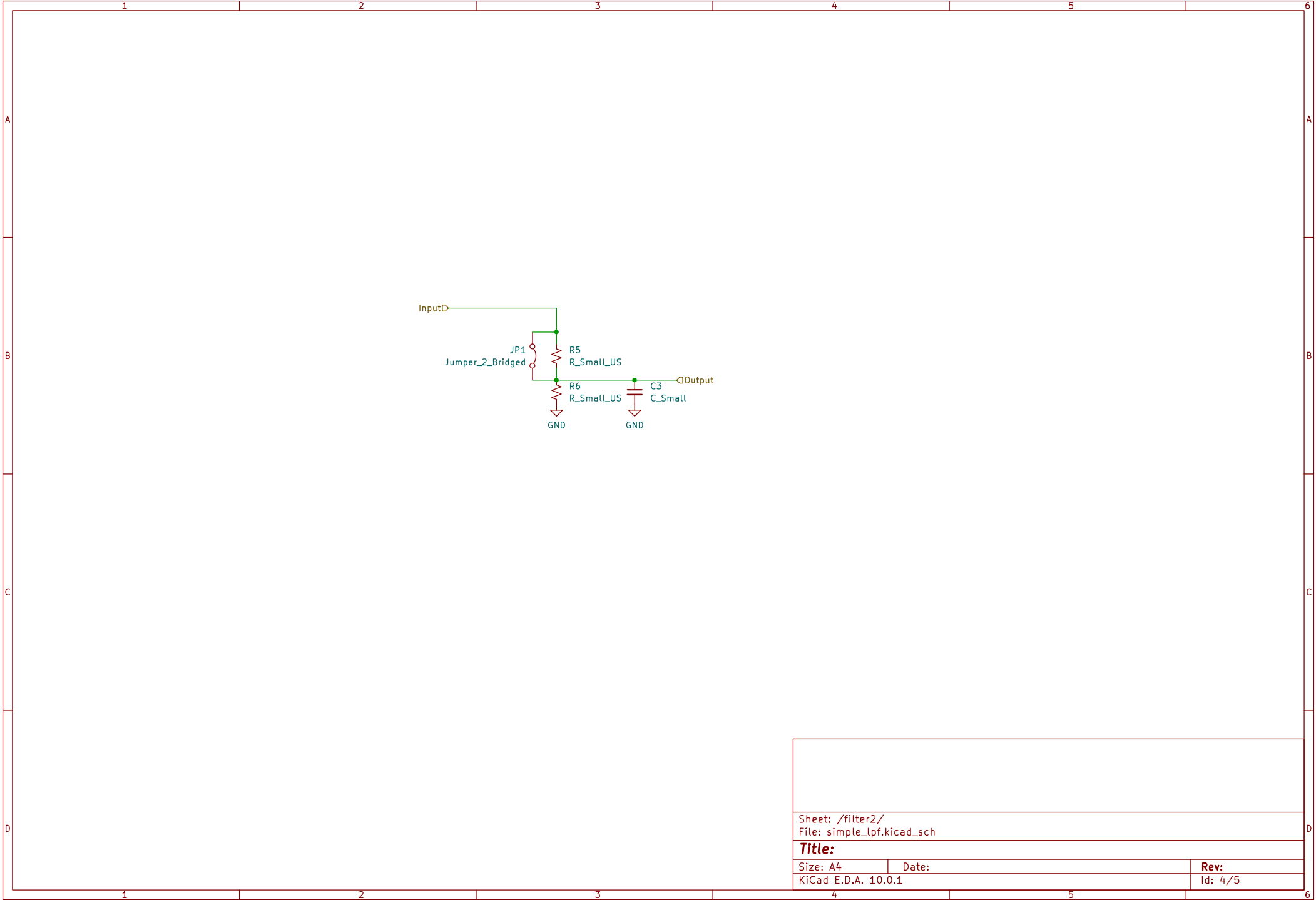
XSHUT4 1
I2C1_SDA/PB9 2
I2C1_SCL/PB8 3
GND 4
+3V3 5

A

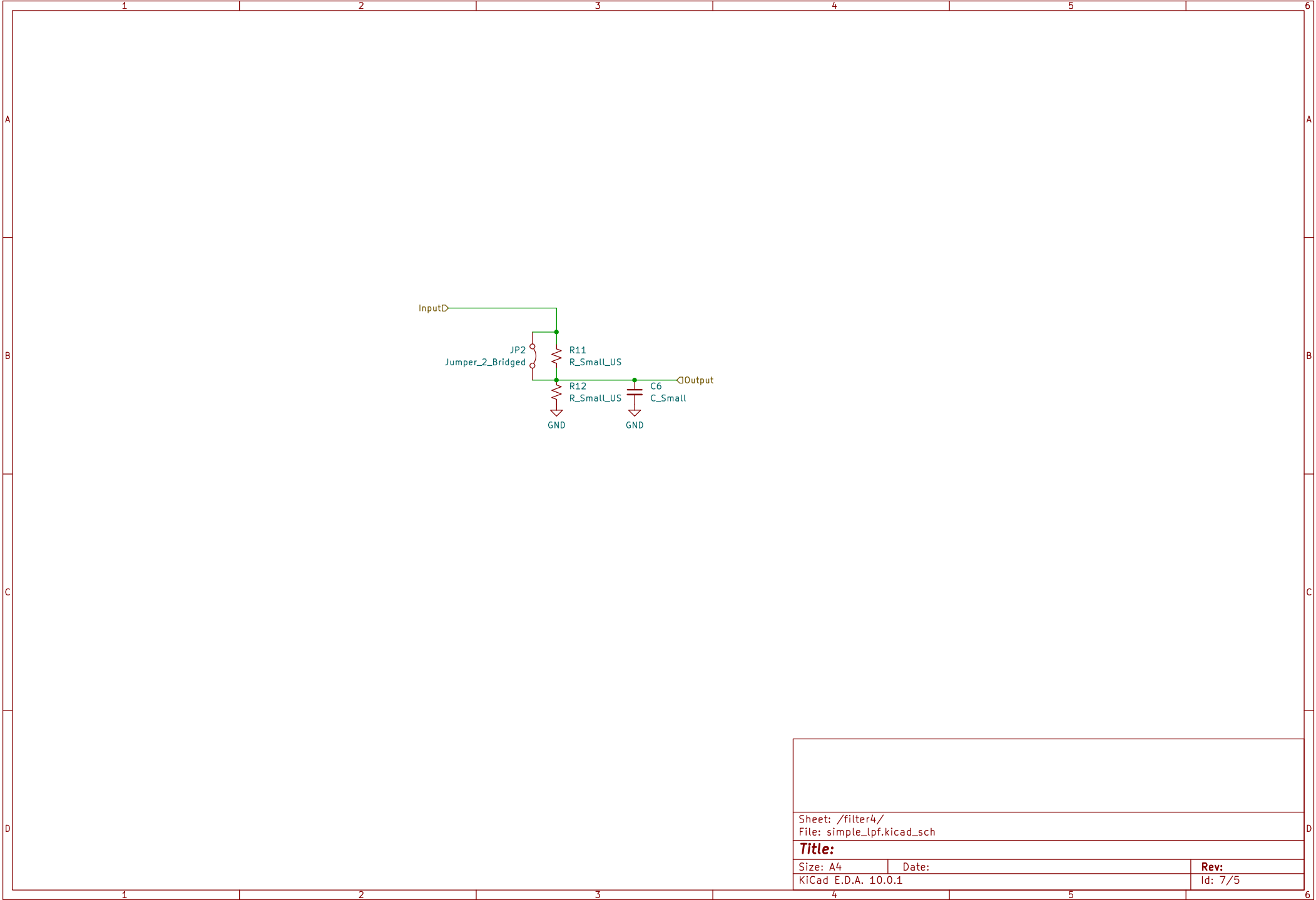
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+3V3 +3V3u

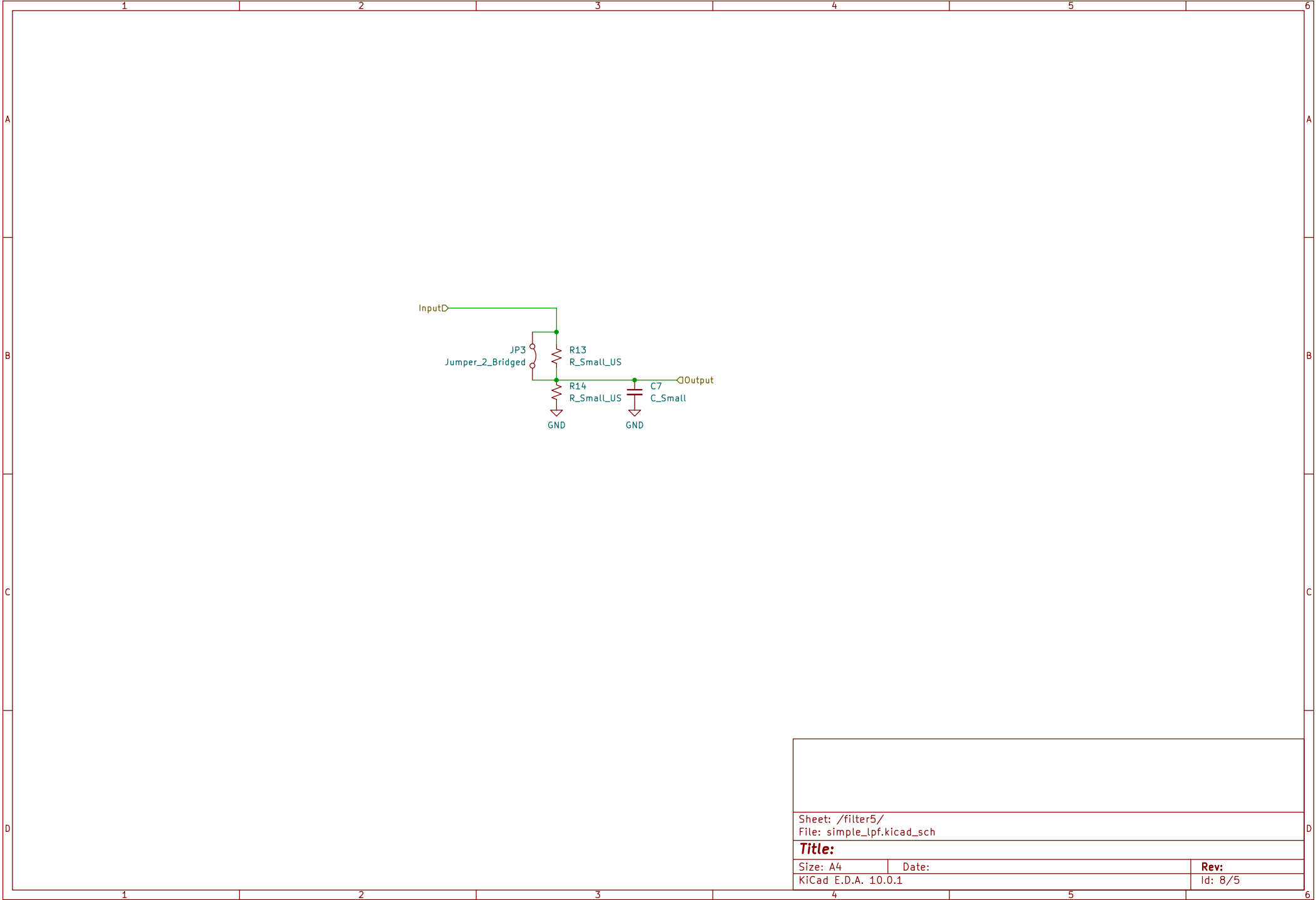
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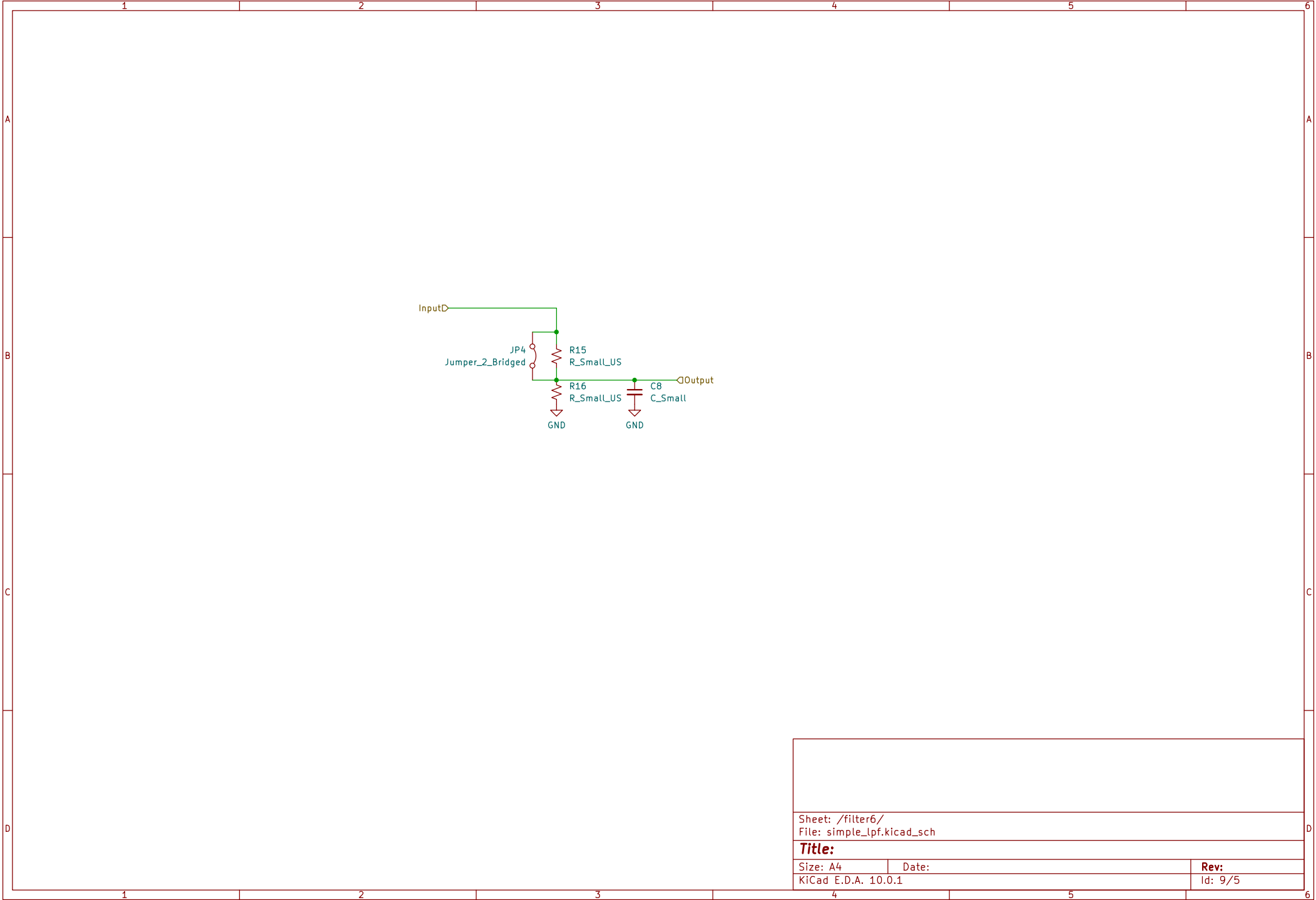
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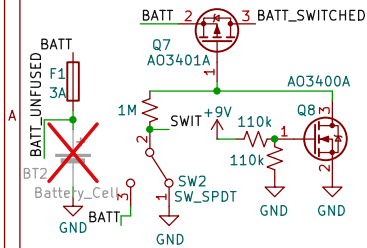
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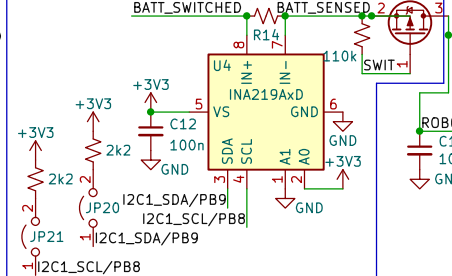




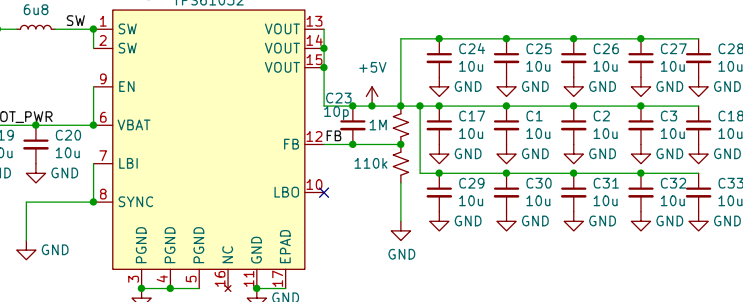
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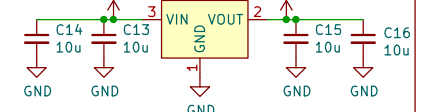
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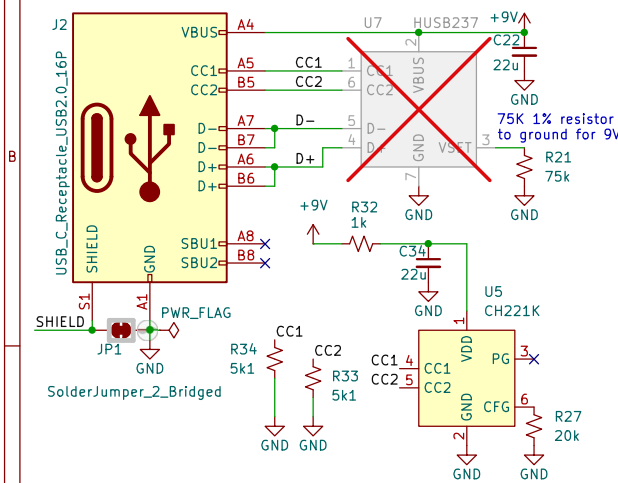
5V Reg



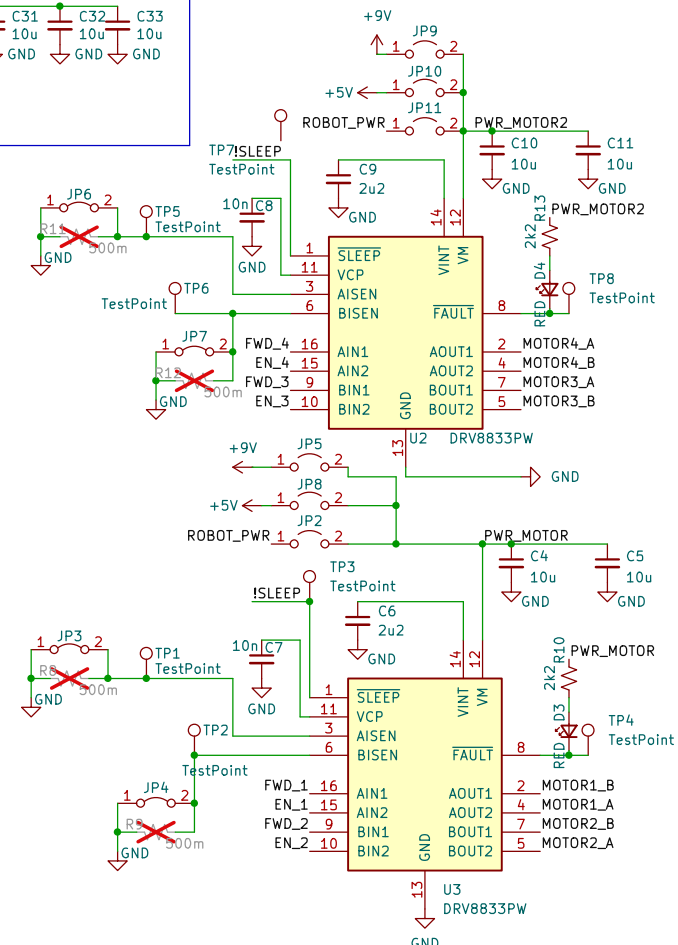
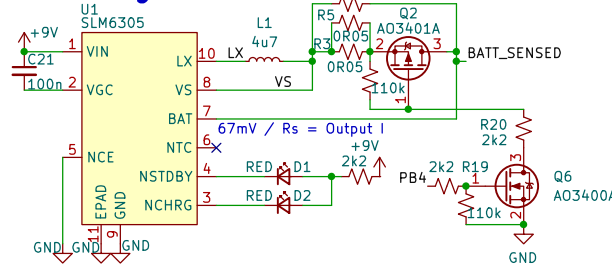
3V3 Reg



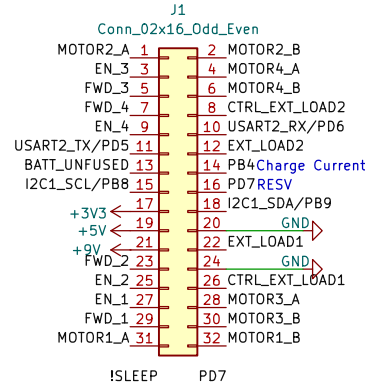
USB C



Li Ion Charger

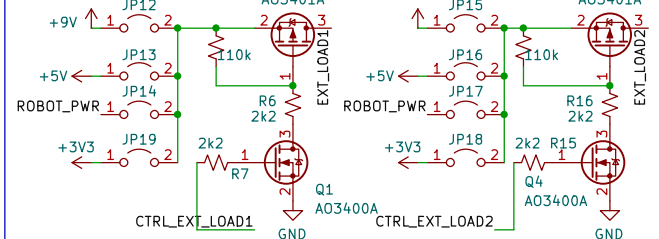


Power Connector



Motor Drivers (4)

Ext Load



Changelog

v0.01 – v0.02
Changed the Motor Driver to run comfortable at a lower voltage
Changed the Mechanical to make the PCB smaller

v0.2 – v1.0
Major rework done.
Changed the Connector to a 2x16
USB C input for 9V (Managed Connection)
Variable Charge current
5V out
3V3 out
Battery Pin for external batteries
Voltage and Current Monitoring (I2C)
Switch for Enabling the robot
4x H Bridges apposed to 2. Variable Input Voltage
2x External Load Control

v1 – v1.1
Added Pullups for I2C
Added Pullups for the Ext Loads

v1.1 – v1.2
Moved the PWM for Motor 1 and 2 to one side
Added the feature to allow the battery to charge with the mouse 'off'.

v1.2 – v1.3
Needed to change the PD IC as it had low stock

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Rev: v1.3

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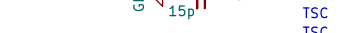
J6
USB_C_Receptacle_USB2.0_16P



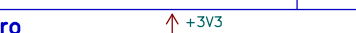
A circuit diagram showing two 3V3 voltage sources connected in series to a 5V output terminal. The output is labeled 5V.



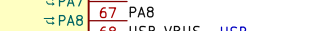
+3V3 R2 +3.3VADC GND



C13



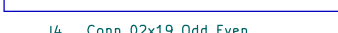
ON0	ON1
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J5 Conn_02x19_Odd_Even



GPIO GPIO GPIO



JP2



v0.1 – v0.2
Changed the programming connector to have 5V present on it – QU

v0.2 – v0.3
Tied the AD0 pin to high on the IMU

v0.3 – v0.4
Added the Y

v0.4 - v0.5
Changed out the MPU6050 for the ICM-42605

v0.5 - v0.6

Added a 100nF on NRST
trace optimisations

v0.6 - v0.7
Added Dodgy flash
Changed the IMI again

v0.7 - v0.8
Fixed incorrect flash footprint

v0.8 – v0.9
Added an LFE on BC14, and BC

- Added an LSE on PC14 and PC15
- Changed LED1 and LED2 onto Analog pins
- Added a testpoint to PC12
- Added USB-C

Added USB_C
Moved USB_ID to a TP

By: Heather Wimberley (v0.1)

by: Heather Wimbereley (VO.1)

@Checked By: Heather Wimberley (v0.1)

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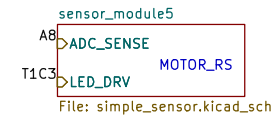
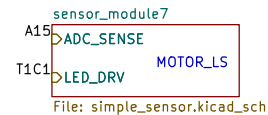
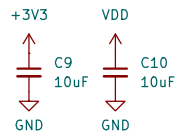
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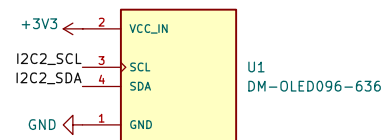
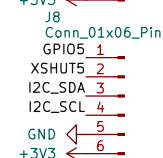
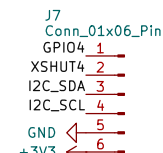
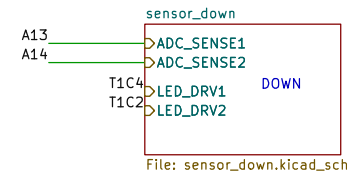
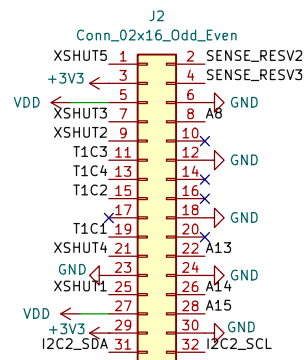
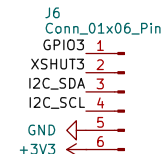
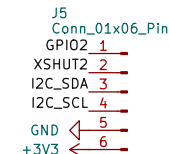
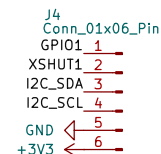
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Page 10 of 10



Changelog
v0.1 – 0.2
Changed the Photodiode to operate on 3V3 and the LEDs to operate on Battery.
v0.2 – 0.3
Changed the Mechanical layout and sensor placement
v0.3 – 0.4
Added 2 potential Line following Sensors
Changed the Mechanical layout and sensor placement again.
v0.4 – 1.0
Major Change to include TOF distance sensors and downward facing LEDs
Changed the sensor Connector to a 32 pin instead of 28 pins
Added I2C Screen pinout on the board

I2C2_SDA I2C_SDA
I2C2_SCL I2C_SCL



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@Checked By:

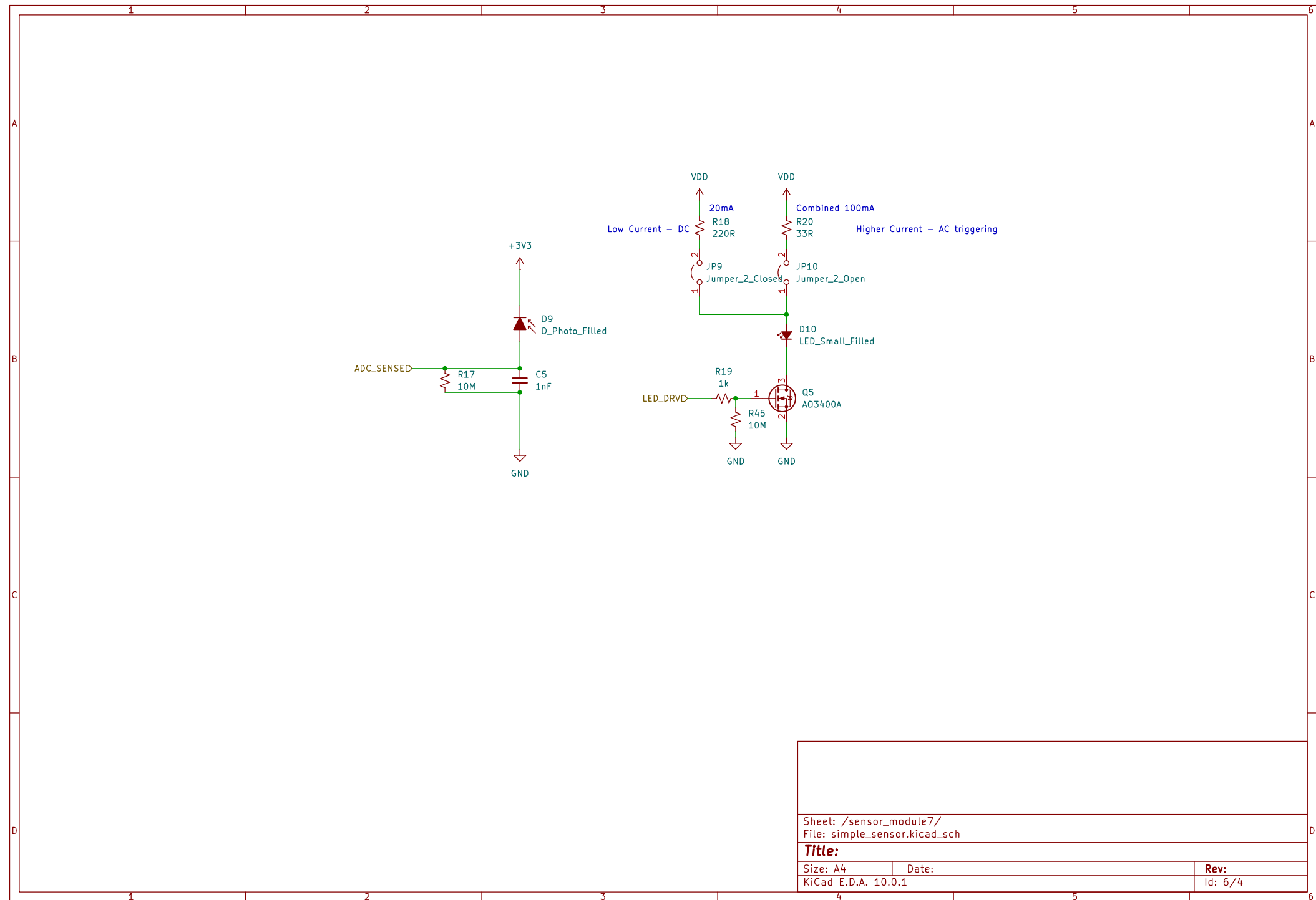
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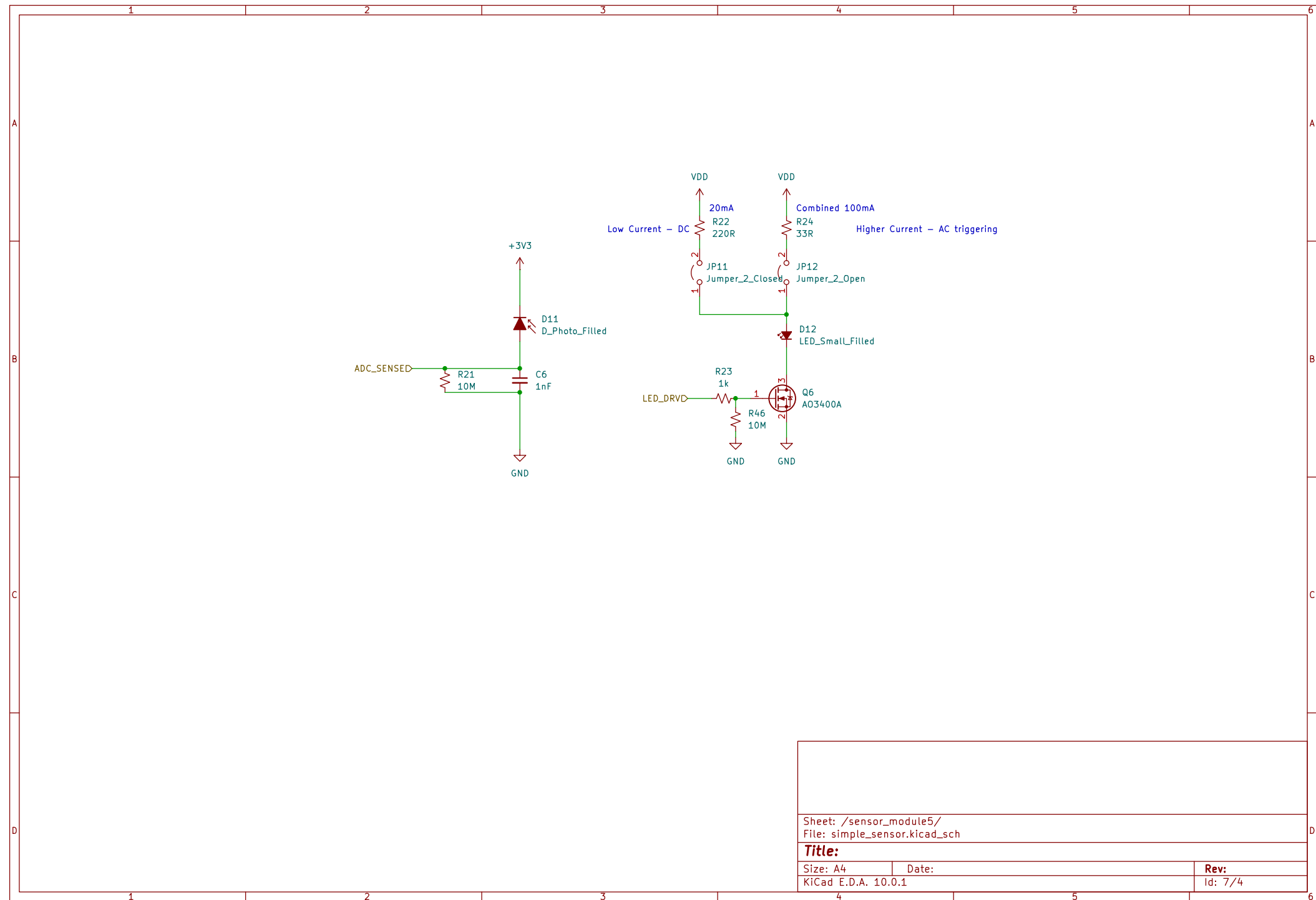
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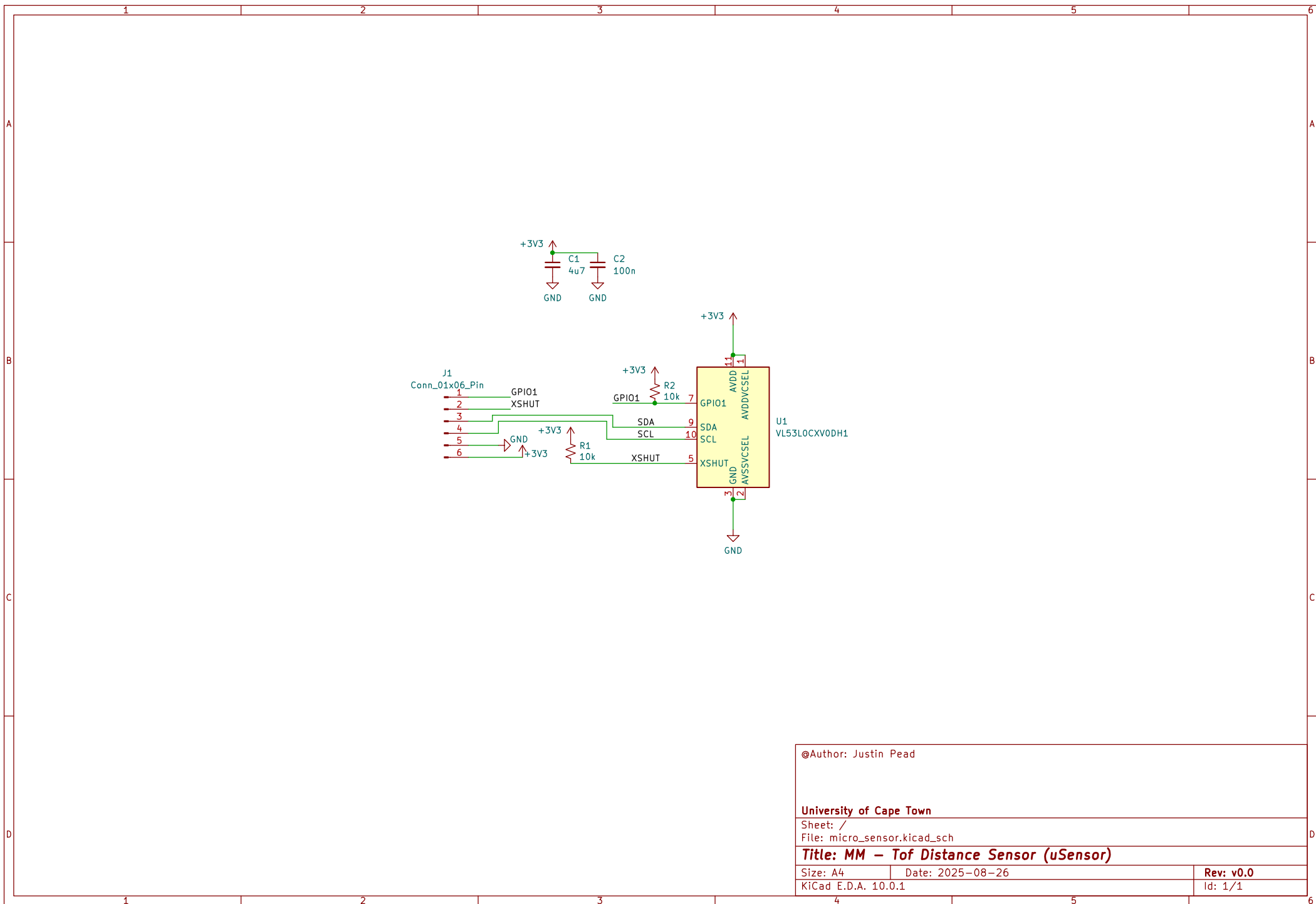
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